

EXPLORATORY DATA ANALYSIS

A Fundamental Pillar of Data Science

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What is Exploratory Data Analysis?

EDA is the process of analyzing datasets to summarize their main characteristics, often using visual and statistical methods, before applying formal modeling techniques.

- 1 Understand structure and patterns hidden in raw data

- 2 Detect anomalies, outliers, and errors early

- 3 Test underlying assumptions before modeling

- 4 Guide feature engineering and model selection

Key Techniques Used in EDA

Descriptive Statistics

Mean, median, mode, standard deviation, and range to summarize central tendency and spread.

Data Visualization

Histograms, box plots, scatter plots, and bar charts reveal distributions and relationships.

Correlation Analysis

Correlation matrices and heatmaps identify relationships between numerical variables.

Missing Value & Outlier Check

Identify nulls, duplicates, and extreme values that could distort analysis.

Typical EDA Process Flow



Each stage feeds the next: clean data enables valid statistics, statistics inform which visuals to build, and visuals surface the insights that guide modeling decisions.

Common Tools & Libraries for EDA

1

Pandas

Data manipulation, summary statistics, and handling missing values in Python.

2

Matplotlib / Seaborn

Static visualizations — histograms, box plots, heatmaps, pair plots.

3

NumPy

Numerical computation underlying statistical operations on arrays.

4

Excel / Power BI

Quick, interactive exploration for smaller or business-facing datasets.

Key Takeaways

- EDA is the foundation of every reliable data science project
- It reveals structure, quality issues, and relationships before modeling begins
- Visual + statistical methods together give the most complete picture
- Skipping EDA risks building models on flawed or misunderstood data

Thank You